

Prompt Ladder

Track: General AI Fluency

Assignment: The Prompt Ladder

Task Chosen: Help frame my Machine Learning task for the FlyRank ML Internship.

Baseline Prompt

1. Prompt

“Help me frame my ML task.”

2. AI Output

Machine learning tasks can be classification, regression, clustering, or ranking. First identify your data, define a target variable, and choose an evaluation metric before building a model.

3. Reflection

- **What changed in the prompt?**
Nothing. This is the original one line prompt I would have written before learning prompt engineering.
- **What improved in the output?**
This is the baseline, so there was nothing to compare yet.
- **What still failed?**
The response was very generic and explained machine learning concepts instead of helping with my internship project. It didn't know what I was building or what assignment I was working on.
- **What I would try next**
Add a clear goal so the AI knows exactly what I need help with.

Version 1 – Clear Goal

1. Technique Added

Clear Goal

2. Prompt

Help me frame my ML task for a Content Refresh Opportunity Scoring project.

3. AI Output

Since your project focuses on Content Refresh Opportunity Scoring, you should first define the decision your model supports, identify the target, and explain why a scoring approach is more suitable than a simple classification model.

4. Reflection

- **What changed in the prompt?**
I added the specific goal of my project instead of simply saying "ML task."
- **What improved in the output?**
The response immediately focused on my actual project rather than giving a generic explanation of machine learning. The suggestions became much more relevant.
- **What still failed?**
The AI still didn't know this was part of the FlyRank internship, so some advice was too broad.
- **What I would try next**
Provide background about the internship and the dataset.

Version 2 – Add Context

1. Technique Added

Context

2. Prompt

Help me frame my ML task for a Content Refresh Opportunity Scoring project. This project is part of the FlyRank ML Internship. I am using the starter dataset to rank pages that may need content updates based on search and engagement signals.

3. AI Output

Based on your internship project, your ML task is a scoring problem where pages receive a priority score instead of a binary label. The project should support content teams by identifying which pages deserve review first.

4. Reflection

- **What changed in the prompt?**
I added the internship context, the dataset, and explained what I wanted to achieve.
- **What improved in the output?**
The AI now understood my project and gave suggestions that matched my assignment instead of generic ML advice. The response became much more useful.
- **What still failed?**
The answer was still presented as one long explanation, making it difficult to copy into my notebook.
- **What I would try next**
Ask for a specific output format.

Version 3 – Specify the Output Format

1. Technique Added

Output Format

2. Prompt

Help me frame my ML task for a Content Refresh Opportunity Scoring project. This project is part of the FlyRank ML Internship using the starter dataset. Organize your answer using the following headings:

- ML Task Type
- Target
- Success Metric
- Unit of Analysis
- Why ML Instead of Rules

3. AI Output

ML Task Type

Scoring / Ranking

Target or Proxy

A refresh priority score representing how urgently a page should be reviewed.

Success Metric

A ranking that is understandable, explainable, and useful for prioritizing content review.

Unit of Analysis

One row represents one content page.

Why ML Instead of Rules

Multiple performance signals interact, making simple thresholds less effective.

4. Reflection

What changed in the prompt?

I requested a structured output using the same headings required by my assignment.

What improved in the output?

The response became much easier to read and I could directly use most of it in my notebook without reorganizing it.

What still failed?

Some explanations were still more technical than necessary for my current level of understanding.

What I would try next

Tell the AI who the audience is so it can adjust the explanation.

Version 4 – Define the Audience

1. Technique Added

Audience

2. Prompt

Help me frame my ML task for a Content Refresh Opportunity Scoring project. This project is part of the FlyRank ML Internship using the starter dataset. Organize your answer using the headings: ML Task Type, Target or Proxy, Success Metric, Unit of Analysis, and Why ML Instead of Rules. Write the explanation for a Computer Science student who is still learning Machine Learning. Avoid unnecessary jargon and keep the language simple.

3. AI Output (Representative Excerpt)

- **ML Task Type**
This project is a scoring task because each page receives a priority score instead of a simple Yes or No label.
- **Target or Proxy**
The target is a refresh priority score that estimates how urgently a page should be reviewed.
- **Why ML Instead of Rules**
A single metric such as CTR or impressions cannot explain page performance by itself. ML can combine several signals to produce a better ranking.

4. Reflection

- **What changed in the prompt?**
I specified who the answer was for.
- **What improved in the output?**
The explanation became much easier to understand. The AI avoided complex terminology and explained the concepts in a way that matched my current knowledge level.
- **What still failed?**
Some examples were generic and didn't directly relate to my internship project.

- **What I would try next**

Ask the AI to verify that every answer matches my actual project and avoid making assumptions.

Version 5 – Add Verification Requirements

1. Technique Added

Verification

2. Prompt

Help me frame my ML task for a Content Refresh Opportunity Scoring project. This project is part of the FlyRank ML Internship using the starter dataset. Organize your answer using the headings: ML Task Type, Target or Proxy, Success Metric, Unit of Analysis, and Why ML Instead of Rules. Write for a Computer Science student using simple language. Before finishing, verify that every section is consistent with the information I provide. Do not invent completed work, results, or experiments.

3. AI Output

ML Task Type

Scoring

Target or Proxy

Refresh Priority Score

Success Metric

A ranking that is understandable, explainable, and supports decision-making.

Why ML Instead of Rules

The project is still in the exploration phase, so a simple rule-based baseline should be created first before evaluating whether ML provides additional value.

4. Reflection

What changed in the prompt?

I asked the AI to verify its answer and avoid making unsupported assumptions.

What improved in the output?

The response accurately reflected my current project stage instead of pretending I had already built a model. It became more trustworthy and realistic.

What still failed?

The response became slightly longer. Although it was more accurate, some extra explanation was unnecessary.

What I would try next

Keep this version because accuracy is more important than saving a few lines.

Honest Observation

Not every prompt improvement made the output dramatically better. Adding the audience improved readability, but it did not significantly improve the technical accuracy. The biggest improvements came from adding project context and asking the AI to verify its assumptions. This exercise showed me that some prompt techniques have a much larger impact than others depending on the task.

Final Reusable Prompt

I am working on a **[PROJECT NAME]**.

Act as an experienced **[ROLE]**.

Context

[Briefly describe the project, the available data, and the objective.]

Task

Help me complete **[TASK]**.

Output Format

Organize the response using these headings:

- [Heading 1]
- [Heading 2]
- [Heading 3]
- [Heading 4]

Requirements

- Use simple language appropriate for a Computer Science student.
- Base every recommendation only on the information provided.
- Do not invent completed work, experiments, or results.
- Clearly separate completed work from future work.

- If assumptions are necessary, state them explicitly.
- Before finishing, verify that every section is consistent with the provided context.